
ReplicationController and ReplicaSet in Kubernetes

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ReplicationController manifest

```
root@ip-172-31-36-215: /home/ubuntu
kind: ReplicationController
apiVersion: v1
metadata:
  name: myreplica
spec:
  replicas: 2
  selector:
    myname: Rohit
  template:
    metadata:
      labels:
        myname: Rohit
    spec:
      containers:
        - name: c00
          image: ubuntu
          command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
```

1.

EXAMPLE OF REPLICATION CONTROLLER

(This manifest defines a Kubernetes ReplicationController named myreplica that maintains 2 replicas of a Pod running an Ubuntu container.

'ReplicationController' is responsible for ensuring that a specified number of pod replicas are running at any given time.

'replicas: 2' This specifies that the ReplicationController should maintain 2 replicas of the Pod.

If any of the Pods fail or are deleted, the ReplicationController will create new Pods to ensure that there are always 2 running.

'myname: Rohit' This selector specifies that the ReplicationController will manage Pods that have the label myname with the value Rohit.

This is crucial for the ReplicationController to know which Pods to monitor and manage. The label myname: Rohit is applied to the Pods.

This label is important because it matches the selector defined earlier, allowing the ReplicationController to manage these Pods.

ReplicationController: The ReplicationController ensures that there are always 2 Pods running that match the specified label selector (myname: Rohit).

If one of the Pods crashes or is deleted, the ReplicationController will automatically create a new Pod to replace it.

Pod Template: The template section defines how the Pods should be created.

Each Pod will run a single container based on the Ubuntu image, executing a command that prints "Hello-Rohit" every 5 seconds.)

```
kind: ReplicationController
```

```
apiVersion: v1
```

```
metadata:
```

```
name: myreplica
```

```
spec:
```

```
replicas: 2
```

```
selector:
```

```
myname: Rohit
```

```
template:
```

```
metadata:
```

```
labels:
```

```
myname: Rohit
```

```
spec:
```

```
containers:
```

```
- name: c00
```

```
image: ubuntu
```

```
command: ["/bin/bash", "-c", "while true; do echo Hello-Rohit; sleep 5; done"]
```

How to list the "ReplicationControllers"

```
root@ip-172-31-36-215:/home/ubuntu# vi myreplica.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f myreplica.yml
replicationcontroller/myreplica created
root@ip-172-31-36-215:/home/ubuntu# kubectl get replicationcontrollers
NAME          DESIRED    CURRENT    READY    AGE
myreplica     2          2          2        59s
root@ip-172-31-36-215:/home/ubuntu# kubectl get rc
NAME          DESIRED    CURRENT    READY    AGE
myreplica     2          2          2        69s
root@ip-172-31-36-215:/home/ubuntu#
```

2.
kubectl get replicationcontrollers
or
kubectl get rc (shortform)

How to check details of the RC

```
root@ip-172-31-36-215:/home/ubuntu# kubectl describe rc myreplica
Name:          myreplica
Namespace:     default
Selector:      myname=Rohit
Labels:        myname=Rohit
Annotations:   <none>
Replicas:      2 current / 2 desired
Pods Status:   2 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  myname=Rohit
  Containers:
    c00:
      Image:        ubuntu
      Port:          <none>
      Host Port:     <none>
      Command:
        /bin/bash
        -c
        while true; do echo Hello-Rohit; sleep 5; done
      Environment:  <none>
      Mounts:        <none>
      Volumes:        <none>
      Node-Selectors: <none>
      Tolerations:    <none>
Events:
  Type      Reason            Age   From
  ----      -
  Normal    SuccessfulCreate   2m8s  replication-controller
  Normal    SuccessfulCreate   2m8s  replication-controller
Created pod: myreplica-8ptzq
Created pod: myreplica-9zcl6
root@ip-172-31-36-215:/home/ubuntu#
```

3.
kubectl describe rc rc_name

ReplicationController maintains the desired state

```

root@ip-172-31-36-215: /home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myreplica-8ptzq                     1/1     Running   0           4m53s
myreplica-9zcl6                     1/1     Running   0           4m52s
testpod3                            3/3     Running   0 (24m ago)  16h
root@ip-172-31-36-215: /home/ubuntu# kubectl delete pod myreplica-8ptzq
pod "myreplica-8ptzq" deleted
root@ip-172-31-36-215: /home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
myreplica-2929f                     1/1     Running   0           38s
myreplica-9zcl6                     1/1     Running   0           5m52s
testpod3                            3/3     Running   3 (25m ago)  16h
root@ip-172-31-36-215: /home/ubuntu# kubectl get rc
NAME            DESIRED   CURRENT   READY   AGE
myreplica       2         2         2       6m24s
root@ip-172-31-36-215: /home/ubuntu#

```

4.

EVEN IF WE DELETE A POD IT'LL CREATE A NEW POD RIGHT AWAY AS THE DESIRED STATE IS 2 REPLICAS

How to check the logs(How many pods were created and deleted)

```

Name:          myreplica
Namespace:     default
Selector:      myname=Rohit
Labels:        myname=Rohit
Annotations:   <none>
Replicas:      2 current / 2 desired
Pods Status:   2 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  myname=Rohit
  Containers:
    c00:
      Image:      ubuntu
      Port:       <none>
      Host Port:  <none>
      Command:
        /bin/bash
        -c
        while true; do echo Hello-Rohit; sleep 5; done
  Environment:  <none>
  Mounts:       <none>
  Volumes:      <none>
  Node-Selectors:  <none>
  Tolerations:   <none>
Events:
  Type     Reason            Age   From
  ----     -
  Normal   SuccessfulCreate   7m12s replication-controller Created pod: myreplica-8ptzq
  Normal   SuccessfulCreate   7m12s replication-controller Created pod: myreplica-9zcl6
  Normal   SuccessfulCreate   118s  replication-controller Created pod: myreplica-2929f
root@ip-172-31-36-215: /home/ubuntu#

```

5.

kubectl describe rc rc_name

How to list the pods along with the labels

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME                READY   STATUS    RESTARTS   AGE   LABELS
myreplica-2929f     1/1     Running   0           3m8s   myname=Rohit
myreplica-9zcl6     1/1     Running   0           8m22s   myname=Rohit
testpod3            3/3     Running   3 (28m ago)  16h    <none>
```

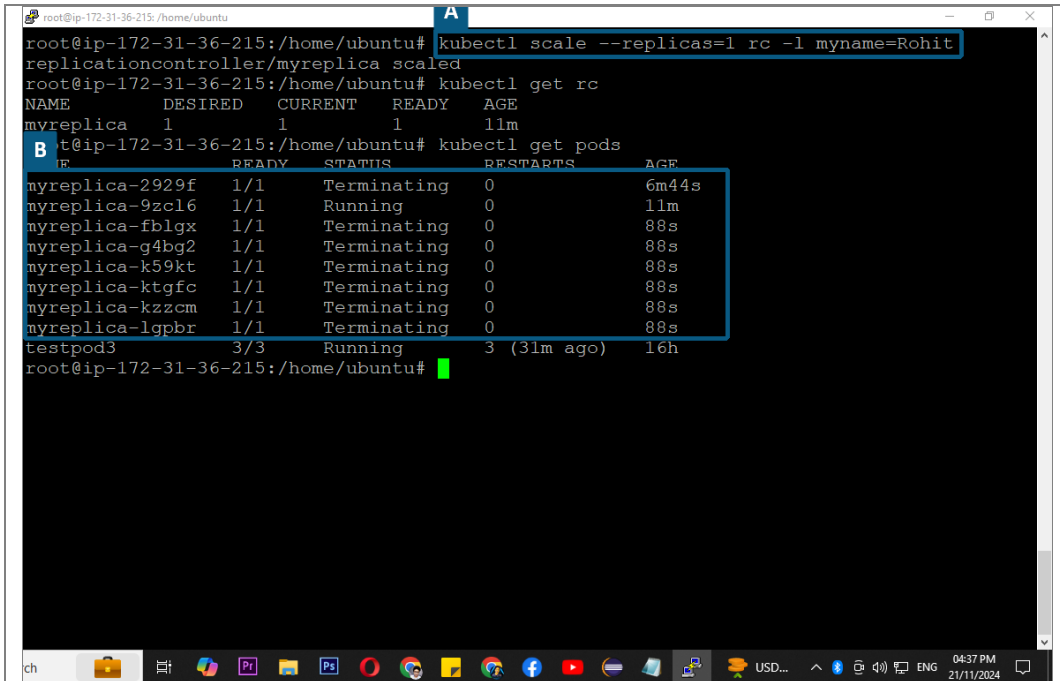
- 6.
- kubectl get pods --show-labels

How to scale up the pods(replicas) in the rc using imperative command

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods --show-labels
NAME                READY   STATUS    RESTARTS   AGE   LABELS
myreplica-2929f     1/1     Running   0           3m8s   myname=Rohit
myreplica-9zcl6     1/1     Running   0           8m22s   myname=Rohit
testpod3            3/3     Running   3 (28m ago)  16h    <none>
root@ip-172-31-36-215:/home/ubuntu# kubectl scale --replicas=8 rc -l myname=Rohit
replicationcontroller/myreplica scaled
root@ip-172-31-36-215:/home/ubuntu# kubectl get rc
NAME                DESIRED   CURRENT   READY   AGE
myreplica           8          8          8       10m
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
myreplica-2929f     1/1     Running   0           5m36s
myreplica-9zcl6     1/1     Running   0           10m
myreplica-fblgx     1/1     Running   0           20s
myreplica-g4bg2     1/1     Running   0           20s
myreplica-k59kt     1/1     Running   0           20s
myreplica-ktgfc     1/1     Running   0           20s
myreplica-kzzcm     1/1     Running   0           20s
myreplica-lqpbr     1/1     Running   0           20s
testpod3            3/3     Running   3 (30m ago)  16h
```

- 7.
- kubectl scale --replicas=*desired number* rc -l key=value(label)
- kubectl scale --replicas=8 rc -l myname=Rohit

How to scale down the pods(replicas) in the rc using imperative command



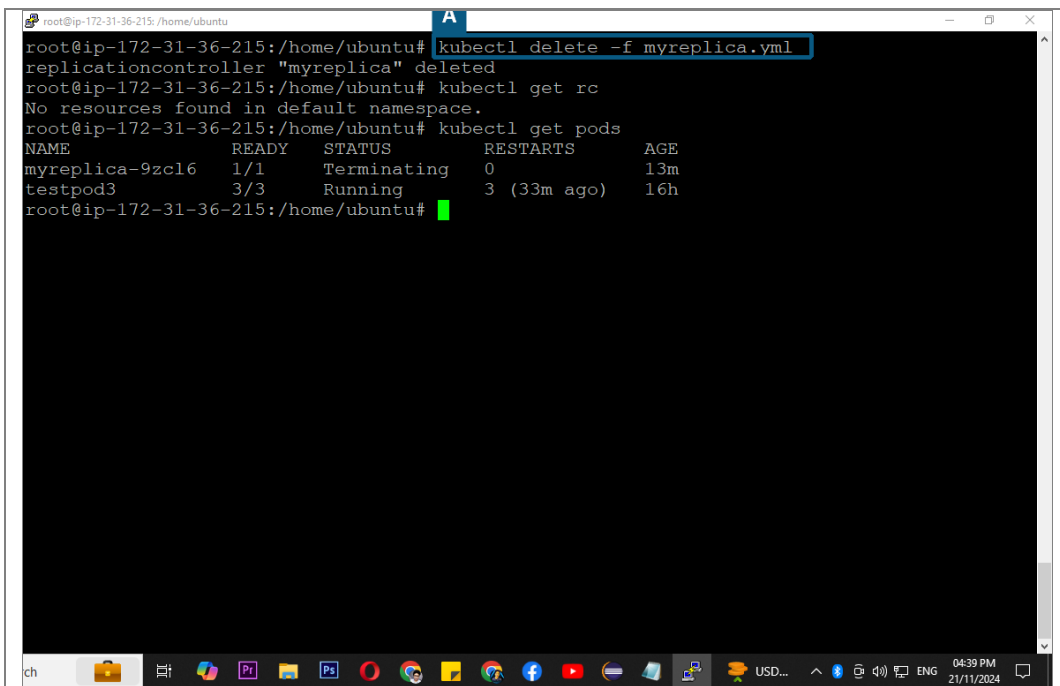
```
root@ip-172-31-36-215:/home/ubuntu# kubectl scale --replicas=1 rc -l myname=Rohit
replicationcontroller/myreplica scaled
root@ip-172-31-36-215:/home/ubuntu# kubectl get rc
NAME           DESIRED   CURRENT   READY   AGE
myreplica       1          1          1       11m
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
myreplica-2929f     1/1     Terminating    0           6m44s
myreplica-9zcl6     1/1     Running         0           11m
myreplica-fblgx     1/1     Terminating    0           88s
myreplica-g4bg2     1/1     Terminating    0           88s
myreplica-k59kt     1/1     Terminating    0           88s
myreplica-ktgfc     1/1     Terminating    0           88s
myreplica-kzzcm     1/1     Terminating    0           88s
myreplica-lgpbr     1/1     Terminating    0           88s
testpod3            3/3     Running         3 (31m ago)  16h
root@ip-172-31-36-215:/home/ubuntu#
```

8.

`kubectl scale --replicas=*desired number* rc -l key=value(label)`

`kubectl scale --replicas=1 rc -l myname=Rohit`

How to delete the ReplicationController (delete the yaml file)



```
root@ip-172-31-36-215:/home/ubuntu# kubectl delete -f myreplica.yml
replicationcontroller "myreplica" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get rc
No resources found in default namespace.
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
myreplica-9zcl6     1/1     Terminating    0           13m
testpod3            3/3     Running         3 (33m ago)  16h
root@ip-172-31-36-215:/home/ubuntu#
```

9.

`kubectl delete -f rcfilename.yml`

ReplicaSet manifest

```
root@ip-172-31-36-215: /home/ubuntu
kind: ReplicaSet
apiVersion: apps/v1
metadata:
  name: myrs
spec:
  replicas: 2
  selector:
    matchExpressions:
      - key: myname
        operator: In
        values:
          - Rohit
          - Roheet
          - Ro
      - key: env
        operator: NotIn
        values:
          - production
  template:
    metadata:
      name: testpod7
      labels:
        myname: Rohit
    spec:
      containers:
        - name: c00
          image: ubuntu
          command: ["/bin/bash", "-c", "while true; do echo Bay-Harbour-Butcher; sleep 5
; done"]
-- INSERT --
28,97 All
```

10.

EXAMPLE OF REPLICASET

('ReplicaSet' ensures that a specified number of pod replicas are running at any given time, similar to a 'ReplicationController', but with more advanced features. The 'apps/v1' version is used for resources like 'Deployment', 'ReplicaSet', and 'StatefulSet'.

replicas: 2 This specifies that the ReplicaSet should maintain 2 replicas of the Pod.

selector: This defines how the 'ReplicaSet' identifies the Pods it manages.

matchExpressions: This allows for more complex selection criteria compared to simple equality-based selectors.

It uses logical operators to match labels.

key: myname, operator: In, values: [Rohit, Roheet, Ro]

This means that the ReplicaSet will manage Pods that have the label myname with a value of either Rohit, Roheet, or Ro.

key: env, operator: NotIn, values: [production]

This means that the ReplicaSet will exclude Pods that have the label env with the value production.

In summary, while both ReplicationController and ReplicaSet serve the purpose of maintaining a specified number of Pod replicas, the ReplicaSet offers more advanced features, flexibility in label selection, and is part of the modern Kubernetes API.

The ReplicationController is largely considered outdated and is typically replaced by ReplicaSet or Deployment in contemporary Kubernetes practices.)

kind: ReplicaSet

apiVersion: apps/v1

metadata:

name: myrs

spec:

replicas: 2

selector:

matchExpressions:

- key: myname

operator: In

values:

- Rohit

- Roheet

- Ro

- key: env

operator: NotIn

values:

- production

template:

metadata:

name: testpod7

labels:

myname: Rohit

spec:

containers:

- name: c00

image: ubuntu

command: ["/bin/bash", "-c", "while true; do echo Bay-Harbour-Butcher; sleep 5; done"]

How to list the RS

```
root@ip-172-31-36-215:/home/ubuntu# vi myrs.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f myrs.yml
replicaset.apps/myrs created
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME          DESIRED  CURRENT  READY  AGE
myrs          2         2         2      2m6s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY  STATUS   RESTARTS  AGE
myrs-8gnp4    1/1    Running  0          2m18s
myrs-d59bj    1/1    Running  0          2m18s
testpod3      3/3    Running  3 (66m ago)  16h
root@ip-172-31-36-215:/home/ubuntu#
```

11.

Scaling down the replicas number from 2 to 1

```
root@ip-172-31-36-215:/home/ubuntu# vi myrs.yml
root@ip-172-31-36-215:/home/ubuntu# kubectl apply -f myrs.yml
replicaset.apps/myrs created
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME          DESIRED  CURRENT  READY  AGE
myrs          2         2         2      2m6s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY  STATUS   RESTARTS  AGE
myrs-8gnp4    1/1    Running  0          2m18s
myrs-d59bj    1/1    Running  0          2m18s
testpod3      3/3    Running  3 (66m ago)  16h
root@ip-172-31-36-215:/home/ubuntu# kubectl scale --replicas=1 rs/myrs
replicaset.apps/myrs scaled
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME          DESIRED  CURRENT  READY  AGE
myrs          1         1         1      3m9s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY  STATUS   RESTARTS  AGE
myrs-8gnp4    1/1    Terminating  0          3m13s
myrs-d59bj    1/1    Running      0          3m13s
testpod3      3/3    Running      3 (67m ago)  16h
root@ip-172-31-36-215:/home/ubuntu#
```

12.

```
kubectl scale --replicas=1 rs/rs_name
kubectl scale --replicas=1 rs/myrs
```

ReplicaSet will maintain the desired state

```
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myrs-d59bj    1/1     Running   0           3m53s
testpod3      3/3     Running   3 (6m ago)  16h
root@ip-172-31-36-215:/home/ubuntu# kubectl delete pod myrs-d59bj
pod "myrs-d59bj" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
NAME      DESIRED   CURRENT   READY   AGE
myrs      1         1         1       4m56s
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myrs-ch4lv    1/1     Running   0           49s
testpod3      3/3     Running   3 (69m ago)  16h
root@ip-172-31-36-215:/home/ubuntu#
```

13.

EVEN IF WE DELETE THE POD IT'LL CREATE A NEW ONE RIGHT AWAY

How to delete the ReplicaSet

```
root@ip-172-31-36-215:/home/ubuntu# kubectl delete rs/myrs
replicaset.apps "myrs" deleted
root@ip-172-31-36-215:/home/ubuntu# kubectl get rs
No resources found in default namespace.
root@ip-172-31-36-215:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myrs-ch4lv    1/1     Terminating   0           3m11s
testpod3      3/3     Running        3 (71m ago)  16h
root@ip-172-31-36-215:/home/ubuntu#
```

14.

```
kubectl delete rs/rs_name
kubectl delete rs/myrs
```